Technology Stack

Date	10/07/2026
Name	Karuru Arjun
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, AI models, audio-processing libraries, front-end tools, development environment, and third-party libraries used to build the Voice-Based Concept Understanding Analyser (VBCUA). A suitable technology stack ensures the application is scalable, maintainable, efficient, and capable of providing accurate AI-based speech evaluation.

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	Streamlit, HTML, CSS	Provides a simple, responsive, and interactive user interface for uploading/recording audio, displaying transcripts, evaluation scores, waveform visualizations, and AI feedback.
2	**Backend / Server-Side**	Python 3	Implements the core application logic, integrates AI models, processes uploaded audio, performs semantic analysis, and generates evaluation reports.
3	**AI & Machine Learning Models**	OpenAI Whisper, Sentence-BERT	Whisper converts speech into text with high accuracy. Sentence-BERT computes semantic similarity between the user's explanation and the reference concept to evaluate conceptual understanding.
4	**Audio Processing**	Librosa, NumPy	Extracts speech features such as duration, RMS energy, pause ratio, and other audio metrics used to evaluate speech fluency and confidence.
5	**Data Storage**	Local File System (Audio, Reports, Reference Data)	Stores uploaded audio files, generated PDF reports, waveform images, and reference concept data required for analysis.
6	**Third-Party APIs / Other**	Sentence-Transformers,	Sentence-Transformers powers semantic analysis, PyTorch runs AI models, ReportLab

	Tools	PyTorch, ReportLab, Matplotlib, Pandas	generates PDF reports, Matplotlib creates waveform visualizations, and Pandas manages structured evaluation data.